

Finite CONV* Banks for Joint Warped-Pixel Transfer

Addendum to Compact Ordered Nodal Variation

I. Finite CONV* Banks for Joint Warped-Pixel Transfer

Let $Y : \Lambda \rightarrow \mathbb{R}^c$ be a finite Cartesian source field. CONV assigns signed currents to source intervals, synthesizes two Cartesian factor orders, and combines them by a scalar action coordinate. Joint current admission then produces a continuous tensor-quintic control lattice $P(Y)$. This section compiles that construction and its warped-pixel functionals into finite banks. The coefficients carry the admitted source state, the continuous synthesis phase, and the integration geometry.

A. Finite global action banks

Let $\eta_a \in [0, 1]$ be the nodal order coordinate at source site a and $d_M(x, a)$ the distance of the source-conditioned positive-definite metric. At an off-node query x , define

$$h_M(x, a) = d_M(x, a)^{-2}, \quad \beta(x) = \frac{\sum_{a \in \Lambda} h_M(x, a) \eta_a}{\sum_{a \in \Lambda} h_M(x, a)}. \quad (1)$$

The value at a source node is assigned by cardinal extension, $\beta(a) = \eta_a$. For a finite query set, the two sums are applications of one finite matrix to the columns η and $\mathbf{1}$. Positivity gives $\min_a \eta_a \leq \beta(x) \leq \max_a \eta_a$.

Proposition 1 (Global FIR realization). Suppose M is constant and source coordinates are in lattice units. Let m be the indicator of Λ , extended by zero, and let $\phi \in [0, 1]^2$ be an off-node phase. On the finite set of required source and query differences, put

$$k_\phi[r] = ((r + \phi)^T M (r + \phi))^{-1}. \quad (2)$$

With $(k * f)[n] = \sum_a k[n-a]f[a]$, the global action satisfies

$$\beta(n + \phi) = \frac{(k_\phi * (m\eta))[n]}{(k_\phi * m)[n]}. \quad (3)$$

Proof. For a constant metric, $d_M(n + \phi, a)^2 = (n - a + \phi)^T M (n - a + \phi)$. Substitution into (1) gives the numerator and denominator in (3). Zero extension of m includes precisely the finite source domain. The node value follows from the dominant self term as $x \rightarrow a$. $\square$

Each phase therefore determines two finite convolutions with a shared kernel. Zero padding to the full linear-convolution extent permits their Fourier execution. For a spatially varying metric, the corresponding bank is $h_M(n + \phi, a)$ with both indices retained. Metric construction and distance evaluation establish this bank once for a fixed source and query geometry. The bilinear nodal coordinate $\beta^* = I_h \eta$ supplies a separate compact choice.

For either coordinate the factor bank below has the same form, and changing the coordinate gives the identity

$$\mathcal{T}_{\beta^*} Y - \mathcal{T}_\beta Y = (\beta^* - \beta)(\mathcal{T}_{yx} Y - \mathcal{T}_{xy} Y). \quad (4)$$

B. Active-face synthesis and two-dimensional composition

On a unit source interval, write $a_k = K_k Y$ for the five raw currents, $\delta = DY$ for the source difference, and eY for the left endpoint. The endpoint closures are included in K_k, D, e . The integrated Bernstein tails are

$$\tau_k(u) = \sum_{j=k+1}^5 B_j^5(u), \quad 0 \leq k \leq 4. \quad (5)$$

Let J be the support of the admitted current. On a nonempty active face, the Euclidean signed-fibre projection gives

$$c_k = \begin{cases} a_k + \frac{\delta - \sum_{l \in J} a_l}{|J|}, & k \in J, \\ 0, & k \notin J. \end{cases} \quad (6)$$

The ordered ledger and projection determine J from the input line.

Proposition 2 (Face-indexed FIR synthesis). For $J \neq \emptyset$, define

$$\begin{aligned} \bar{\tau}_J(u) &= |J|^{-1} \sum_{k \in J} \tau_k(u), \\ W_J(u) &= e + \bar{\tau}_J(u) D \\ &\quad + \sum_{k \in J} (\tau_k(u) - \bar{\tau}_J(u)) K_k. \end{aligned} \quad (7)$$

Then the admitted factor value equals $W_J(u)Y$. For the empty face, $W_\emptyset(u) = e$.

Proof. Insert (6) into $eY + \sum_k \tau_k(u) c_k$ and collect the coefficients of δ and a_k . If J is empty, all admitted currents vanish and synthesis returns the anchor. $\square$

There are 31 nonempty face families. Each is polynomial in u of degree at most five and has six-tap interior support. Lineage is retained in the choice of family. The factor can consequently be evaluated by one stencil after its face has been established.

For a source cell (i, j) and phase (u, v) , let $W_{i,j+s}^x(u; r)$ be the first-factor stencil on row $j + s$. Let $W_{i,j,u}^y(v; s)$ be the second-factor stencil established from the line of first-factor values at $i + u$.

Theorem 1 (Conditioned two-dimensional factor bank). The horizontal-vertical order has the finite representation

$$K_{i,j,u,v}^{xy}[r,s] = W_{i,j,u}^y(v;s)W_{i,j+s}^x(u;r),$$

$$(\mathcal{T}_{xy}Y)(i+u, j+v) = \sum_{r,s} K_{i,j,u,v}^{xy}[r,s]Y_{i+r,j+s}. \quad (8)$$

Constructing K^{yx} with exchanged axes gives the complete blended bank

$$K_B = (1 - \beta)K^{xy} + \beta K^{yx}. \quad (9)$$

Proof. Apply (7) first to each source row and then to the resulting transverse line. Substitution of the first finite sum into the second gives (8). The convex factor blend gives (9). $\square$

The interior support is six by six. Different source rows may have different first-factor faces, so the bank is generally nonseparable. On an interval of u where the second-factor face is fixed, its coefficients are independent of u ; the first-factor values remain quintic in u . Each order is then bicquintic on the corresponding phase region. With bilinear β^* , the blend has bidegree at most (6,6) on a region where both orders have fixed faces. Ledger-choice and active-face inequalities certify such regions; phase evaluation within a certified region reuses the same polynomial coefficients.

For an endpoint-aligned rational lattice, write the reduced source-coordinate step on each axis as a_d/b_d . The fractional phases belong to the b_d residue classes modulo one. Their Cartesian product supplies at most $b_x b_y$ phase pairs, with boundary closures and admitted face indices retained as bank indices. Continuous query coordinates evaluate the same polynomial families directly.

C. Compilation through joint current admission

The joint atlas uses a fixed fifth-step collocation map followed by shared range and current admission. Let $B(Y)$ be the degree-elevated bilinear baseline, $\tilde{P}(Y)$ the collocation proposal, and $\hat{P}(Y)$ its range projection. Assign each nonvertex control to one shared entity g . If α_g is its simultaneous capacity and τ the common completion parameter, define

$$\lambda_g = \alpha_g + \tau(1 - \alpha_g), \quad P_e = B_e + \lambda_g(\hat{P}_e - B_e), \quad e \in g. \quad (10)$$

Vertex controls retain their source values.

Theorem 2 (Source-state factorization). Let $S(Y)$ record the faces and order coordinates used at collocation nodes, the selected range-bound source indices, and the admitted values of λ_g . For each channel there is a finite matrix $A_{S(Y)}$ such that

$$P(Y) = A_{S(Y)}Y. \quad (11)$$

Proof. With the recorded faces and coordinates, (9) makes each collocation value a linear source stencil. Inverse Bernstein collocation and bilinear degree elevation are fixed linear maps. Range projection selects one of three linear values: the proposal, the selected lower-bound sample, or

the selected upper-bound sample. Equation (10) then combines two linear stencils with recorded scalar coefficients. Assembling the shared entities gives $A_{S(Y)}$. $\square$

For fixed Y , this factorization retains the exact admitted controls and therefore their cardinality, continuity, support-range bounds, and current cones. State discovery consists of line scans, finite projections, support extrema, incident-constraint minima, and the common ray reduction. Once that state is available, a source stencil can be propagated through every linear stage or the intermediate control lattice can be retained. Both evaluate (11).

D. Point and area banks on the admitted atlas

Use source-pixel coordinates on the source rectangle Ω . Let F map target coordinates to source coordinates and let $G = F^{-1}$. Assume the map is nonsingular and has no denominator zero on the domain of integration. Put $Q = G(\Omega)$ and extend the source field by zero outside Q in target coordinates. On a source cell C , write

$$U_C(u, v) = \sum_{i,j=0}^5 P_{C,ij} B_i^5(u) B_j^5(v). \quad (12)$$

Point synthesis uses the continuous phase bank

$$\Psi_{u,v}[i, j] = B_i^5(u) B_j^5(v). \quad (13)$$

For a target basin E of positive area, let $S_C = F(E \cap Q) \cap C$ and $J_G = |\det DG|$. Define the cell moment bank

$$M_{C,ij}(E, G) = \int_{S_C} B_i^5(u) B_j^5(v) J_G(u, v) du dv. \quad (14)$$

A change of variables gives

$$(\mathcal{B}_F Y)_E = \frac{1}{|E|} \sum_{C,ij} M_{C,ij} P_{C,ij}. \quad (15)$$

After assembly of shared controls and division by $|E|$, denote this bank by $\mathbf{M}_E$. The complete source-to-output operators are

$$(\mathcal{W}_F Y)(q) = \Psi_{F(q)}^T A_{S(Y)} Y, \quad (\mathcal{B}_F Y)_E = \mathbf{M}_E^T A_{S(Y)} Y. \quad (16)$$

Proposition 3 (Positivity and conservation). The exact moment weights are nonnegative and satisfy

$$\sum_{C,ij} M_{C,ij} = |E \cap Q|. \quad (17)$$

For any basin partition $\{E_b\}$,

$$\sum_b |E_b| (\mathcal{B}_F Y)_{E_b} = \int_{(\cup_b E_b) \cap Q} U_Y(F(q)) dq. \quad (18)$$

Proof. The Bernstein basis and J_G are nonnegative. Summing the basis gives one on each source cell. The cell footprints partition $F(E \cap Q)$ up to boundaries of zero area, and change of variables gives $|E \cap Q|$. Summation over the target partition proves the second identity. $\square$

For numerical moment construction, a constant reference R can be integrated through target coverage:

$$(\mathcal{B}_F Y)_E = \frac{R|E \cap Q|}{|E|} + \frac{1}{|E|} \sum_{C,i,j} (P_{C,ij} - R) M_{C,ij}. \quad (19)$$

This identity holds for any R . A reference chosen by a fixed linear functional of the controls preserves the assembled linear-bank form when approximate moments are substituted.

E. Exact affine moment banks

For a rectangular footprint $[a, b] \times [c, d]$ in a cell and constant density ρ , define

$$A_i(t) = \frac{1}{6} \sum_{j=i+1}^6 B_j^6(t), \quad 0 \leq i \leq 5. \quad (20)$$

The Bernstein derivative identity telescopes to $A'_i = B_i^5$, yielding

$$M_{C,ij} = \rho[A_i(b) - A_i(a)][A_j(d) - A_j(c)]. \quad (21)$$

For a general affine map the cell footprint is polygonal and the integrand is polynomial of total degree at most ten. Its monomial moments can be obtained by an oriented boundary sum:

$$\int_D x^a y^b dx dy = \frac{1}{a+1} \oint_{\partial D} x^{a+1} y^b dy. \quad (22)$$

On each line segment, affine parameterization reduces the integrand to a polynomial in one variable. Equations (21) and (22) thus determine finite exact affine banks directly from footprint vertices.

F. Exact projective moment banks

Write G in homogeneous coordinates with nonsingular matrix H_G and affine denominator $\ell(x, y)$. Its area density is

$$J_G(x, y) = \frac{|\det H_G|}{|\ell(x, y)|^3}. \quad (23)$$

Choose a nonzero scale c_0 of the same sign as ℓ on the footprint, and write $\ell = c_0 t$, where $t = \gamma + \alpha x + \beta y > 0$. Let $\rho = |\det H_G|/|c_0|^3$.

Theorem 3 (Finite projective edge compilation). Every entry of (14) on a polygonal cell footprint is a finite expression in its vertices, the map coefficients, rational operations, and logarithms of positive denominator values.

Proof. For $\alpha \neq 0$, transform to (t, y) so that $x = (t - \gamma - \beta y)/\alpha$ and $dx dy = |\alpha|^{-1} dt dy$. The transformed polygon D is given counterclockwise orientation. Expand one tensor Bernstein numerator after substitution as $\sum_{k,l} b_{kl} t^k y^l$, with $k + l \leq 10$. Its bank entry is

$$M = \frac{\rho}{|\alpha|} \sum_{k,l} b_{kl} I_{kl}(D), \quad I_{kl}(D) = \int_D t^{k-3} y^l dt dy. \quad (24)$$

Green's theorem gives

$$I_{kl}(D) = -\frac{1}{l+1} \oint_{\partial D} t^{k-3} y^{l+1} dt. \quad (25)$$

For an oriented edge from (t_0, y_0) to (t_1, y_1) with $t_0 \neq t_1$, put $a = (y_1 - y_0)/(t_1 - t_0)$ and $b = y_0 - at_0$. Its contribution is

$$-\frac{1}{l+1} \sum_{j=0}^{l+1} \binom{l+1}{j} a^j b^{l+1-j} \mathcal{L}_{k-3+j}(t_0, t_1), \quad (26)$$

where

$$\mathcal{L}_e(a, b) = \begin{cases} (b^{e+1} - a^{e+1})/(e+1), & e \neq -1, \\ \log(b/a), & e = -1. \end{cases} \quad (27)$$

An edge with $t_0 = t_1$ contributes zero. If $\alpha = 0$ and $\beta \neq 0$, exchange the source axes. If both vanish, the density is constant and (22) applies. $\square$

The same edge moments serve all 36 basis numerators and every channel. Boundary terms on oppositely oriented shared edges cancel when their chosen Green primitives have the same edge trace. An oriented edge table can therefore reuse primitives across neighboring footprints while preserving each cell's polynomial coefficients.

For close positive a, b , set $h = (b - a)/a$. The primitive difference has the continuous form

$$\mathcal{L}_e(a, b) = \begin{cases} \frac{a^{e+1}}{e+1} \expm1((e+1) \log p(h)), & e \neq -1, \\ \log p(h), & e = -1. \end{cases} \quad (28)$$

This evaluates the power and logarithm differences without subtracting their nearly equal endpoint values. The factors $a^j b^{l+1-j}$ in (26) still require control of cancellation for nearly constant-denominator edges. Scaled cell coordinates, selection of the larger of $|\alpha|, |\beta|$, and the bounded series construction below provide a continuous evaluation across these configurations and the affine limit.

G. Bounded polynomial compilation

Triangulate the clipped cell footprints. On a triangle T , let $[\ell_-, \ell_+]$ be the same-sign denominator range and put

$$c_T = (\ell_- + \ell_+)/2, \quad z = \ell/c_T - 1, \quad r_T = \max_T |z| < 1. \quad (29)$$

The extrema occur at vertices. With $\rho_T = |\det H_G|/|c_T|^3$,

$$J_G = \rho_T (1+z)^{-3}, \quad S_K(z) = \sum_{k=0}^K (-1)^k \binom{k+2}{2} z^k. \quad (30)$$

For $n \geq 1$, the absolute series tail is

$$T_n(r) = \sum_{k=n}^{\infty} \binom{k+2}{2} r^k = \frac{r^n}{2} \left[\frac{r(1+r)}{(1-r)^3} + \frac{(2n+3)r}{(1-r)^2} + \frac{(n+1)(n+2)}{1-r} \right]. \quad (31)$$

The closed form follows by writing $k = n + j$ and summing $1, j, j^2$ against r^j . In particular, $|(1+z)^{-3} - S_K(z)| \leq T_{K+1}(r_T)$.

Proposition 4 (Pixel error certificate). Let $D_{C,a} = \max_{i,j} |P_{C,ij,a} - R_a|$. Replacing J_G by $\rho_T S_K(z)$ on $T \subset C$ in (19) gives the channelwise unnormalized bound

$$E_{T,a} = |T| \rho_T D_{C,a} T_{K+1}(r_T). \quad (32)$$

If the triangle budgets satisfy $\sum_T \epsilon_T \leq \epsilon |E|$ and $\max_a E_{T,a} \leq \epsilon_T$, the normalized pixel error is at most ϵ in every channel.

Proof. Bernstein convexity bounds $|U_C - R_a|$ by $D_{C,a}$. Multiply by the uniform density-tail bound, integrate over T , sum the triangle errors, and divide by $|E|$. $\square$

For a nonempty footprint, one admissible budget is $\epsilon_T = \epsilon |E| |T| / |F(E \cap Q)|$. Four-way subdivision gives each child one quarter of its parent's budget. For any fixed $r_T < 1$ the tail converges to zero as $K \rightarrow \infty$; alternatively subdivision reduces the relative denominator variation. A finite degree or depth cap returns the sum of the attained bounds.

The truncated integrand has total degree at most $10 + K$. A Duffy map to the unit square raises one separate degree to at most $11 + K$. Tensor Gauss order

$$q_K = \left\lceil \frac{12 + K}{2} \right\rceil \quad (33)$$

therefore integrates that polynomial exactly. Polynomial edge moments from (22) give another finite evaluation of the same bank. Affine density has $r_T = 0$ and zero tail. Constant fields have $D_{C,a} = 0$ and are reproduced by exact target coverage.

For each triangle, the geometric moments can be formed before any channel contraction. If p_l, w_l denote quadrature nodes and density-weighted quadrature weights, then

$$\begin{aligned} M_{ij}^{(K)} &= \sum_{l=1}^L w_l B_i^5(p_{l,x}) B_j^5(p_{l,y}), \\ I_{T,a}^{(K)} &= \sum_{i,j} (P_{C,ij,a} - R_a) M_{ij}^{(K)}. \end{aligned} \quad (34)$$

Exchanging the two finite sums changes $36cL$ weighted control contributions into $36L$ geometry contributions and $36c$ control contributions. The triangle partition, polynomial degree, and truncation certificate are preserved.

H. Composition, reuse, and numerical enclosures

The complete dataflow is

$$Y \mapsto S(Y) \mapsto A_{S(Y)} Y \mapsto \{\Psi_q^T P, \mathbf{M}_E^T P\}. \quad (35)$$

Independent lines within each factor share one raw-current bank. The next factor consumes their synthesized values. Shared-entity capacities are minima over incident constraints; the completion parameter is a global minimum. These associative reductions admit parallel trees, while exact lineage uses a prefix scan. Once P is fixed, target basins and channel contractions can be partitioned independently. Repeated phase pairs reuse synthesis weights;

repeated footprints reuse moment banks; common polygon edges reuse oriented primitives. For N_T triangle pieces, moment-first quadrature has contraction work proportional to $\sum_{T=1}^{N_T} (36L_T + 36c)$, in addition to geometry construction.

The identities above hold in real arithmetic. A numerical enclosure can be assembled from storage, moment, and contraction errors. If stored controls satisfy $|\hat{P}_e - P_e| \leq \delta_e$ and computed unnormalized exact-bank approximations satisfy $|\hat{M}_e - M_e| \leq \mu_e$, then

$$\left| \sum_e \hat{M}_e \hat{P}_e - \sum_e M_e P_e \right| \leq \sum_e |\hat{M}_e| \delta_e + \sum_e \mu_e |P_e|. \quad (36)$$

The final summation contributes its arithmetic enclosure and the result is divided by $|E|$. For reference-subtracted polynomial banks, (32) supplies the density-truncation contribution. Bounds on computed coverage, primitive evaluation, and summation are added at the corresponding stages. Exact-bank positivity gives the sharper storage bound $\max_e \delta_e |E \cap Q| / |E|$ when moments are exact.

Thus global admission, nonlinear factor state, joint control admission, and warped integration each have a finite compilation: a normalized global bank, a face-indexed two-dimensional stencil, a source-state matrix, and a geometry-dependent moment bank. Their composition retains the declared continuous source field and its conservative target functional.